

Augmented Educators and AI: Shaping the Future of Human-AI Collaboration in Learning

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Abstract

As Artificial Intelligence (AI) becomes more deeply embedded in educational settings, it is crucial to explore how it can enhance—rather than replace—the role of educators. This workshop focuses on advancing AI-driven tools that support personalized learning and designing AI systems capable of understanding students’ emotional and cognitive needs. The workshop will explore two main themes: (1) empowering teachers with AI technologies for delivering customized feedback and individualized instruction, and (2) examining ethical and practical considerations for developing empathetic AI that complements the human aspects of teaching. Participants will discuss opportunities and challenges in building effective AI-augmented learning environments, considering ethical concerns such as privacy, equity, and the maintenance of human agency. The workshop aims to unite educators, researchers, and technologists in developing innovative solutions and fostering interdisciplinary dialogue. Key outcomes include establishing best practices for AI integration in education and disseminating research findings that shape the future of human-AI collaboration in teaching.

CCS Concepts

• **Human-centered computing** → **Collaborative interaction**; • **Applied computing** → **Collaborative learning**; • **Computing methodologies** → **Intelligent agents**.

Keywords

Augmented Educators, Human-AI Collaboration, Empathetic AI

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1 Background

The integration of Artificial Intelligence (AI) into various sectors, particularly education, is driving a paradigm shift in teaching and learning processes [1], [5], [7]. AI-powered tools and systems hold immense potential to reshape the educational landscape by enhancing the role of educators and fostering deeper human-AI collaboration within classrooms [12]. Central to this transformation is the concept of augmented educators—teachers who leverage AI to amplify their capabilities [14]. This workshop addresses two critical dimensions of this emerging paradigm: (1) equipping educators with AI tools to deliver personalized feedback effectively, and (2) advancing AI systems to exhibit human-like qualities, such as empathy, adaptability, and contextual understanding, in personalized learning environments.

The need for AI in education is becoming more urgent due to the increasing diversity in student needs, learning styles, and socioeconomic backgrounds [4], [10]. Traditional teaching approaches that rely on standardized curricula and manual feedback are no longer sufficient to meet these varied demands. AI technologies, especially those powered by large language models (LLMs), offer a solution by enabling tailored feedback at scale and providing personalized learning experiences across a range of disciplines, from writing and debate to project-based activities [3], [8], [9], [15]. Additionally, AI’s ability to process large amounts of data from student interactions enables real-time insights, allowing teachers to focus on high-impact tasks such as nurturing creativity, critical thinking, and problem-solving [17].

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However, incorporating AI into education brings its own set of challenges. For AI to truly augment educators, it must move beyond simple automation and data analysis. Teachers need AI tools that not only support their instructional roles but also preserve the human connection in learning [6].

As AI becomes more embedded in classrooms, critical questions arise about how these systems can retain the human touch, especially in providing emotionally meaningful, context-aware feedback[16]. The core issues of this workshop revolve around two key themes: designing AI systems that effectively support educational objectives and advancing AI to develop a more empathetic and human-like presence in learning settings.

1.1 AI-Empowered Educators: Enhancing Personalized Feedback and Teaching Effectiveness

The first crucial element of human-AI collaboration in education centers on how AI can empower educators to deliver more personalized feedback, thereby enhancing teaching effectiveness [2]. AI technologies, including LLMs, Intelligent Tutoring Systems (ITS), and adaptive learning platforms, are transforming personalized education by providing tailored, real-time feedback to students. These systems dynamically adjust content based on individual student performance, offering guidance that aligns with each learner's unique abilities, learning styles, and emotional needs [11].

In subjects such as writing and discussion, for example, AI tools can analyze student submissions and deliver immediate, personalized feedback on aspects like grammar, structure, and argumentation. This allows teachers to redirect their efforts toward more complex educational goals, such as promoting critical thinking, creativity, and deeper engagement. In this role, AI functions as a valuable educational partner, enabling teachers to focus on mentoring and fostering meaningful connections with students. Additionally, AI-driven tools can track student progress continuously, offering data-driven insights that help educators refine their instructional strategies and provide targeted support where it has the greatest impact.

Yet, the deployment of AI in education presents important considerations. AI-powered tools must be designed to scale across diverse learning environments and address a wide range of subject areas. LLM-based feedback systems, while capable of delivering instant, customized responses, must also incorporate mechanisms for human oversight to ensure the feedback is accurate and contextually appropriate. It is essential that teachers remain in control of how AI feedback integrates into the educational process, guiding students not only in factual learning but also in their intellectual and emotional development.

1.2 AI with Human-Like Empathy: Creating Systems that Understand and Support Student Emotions

The second significant challenge in human-AI collaboration in education is developing AI systems capable of demonstrating human-like empathy and emotional understanding. For AI to effectively support teachers, it must extend beyond providing personalized

feedback to recognizing and responding to students' emotional and psychological states. AI should be able to detect emotional cues and deliver empathetic, context-aware feedback [13].

This capability is especially vital in areas like social and emotional learning, where a student's emotional state can deeply impact academic performance and overall well-being. Emerging technologies like emotion recognition are being integrated into educational tools to detect signs of frustration or disengagement in real time. By identifying these cues, AI can help teachers provide timely interventions that keep students engaged.

Achieving this level of empathy in AI requires integrating psychological models to interpret human emotions accurately. This effort involves interdisciplinary collaboration among AI developers, psychologists, and educators to ensure AI systems offer appropriate emotional support. Ethical considerations, such as privacy concerns and the risk of over-reliance on AI for emotional needs, also need to be addressed to ensure AI enhances, rather than replaces, human relationships in education.

The future of human-AI collaboration in education depends on creating AI systems that address both cognitive and emotional aspects of learning. Teachers should be empowered with AI tools that enhance personalized feedback, while AI evolves to support students' emotional needs, providing a holistic learning experience.

This workshop will delve into these two core areas—AI for personalized feedback and AI systems with human-like empathy—to explore how AI can transform the role of teachers in reshaping education. By bringing together educators, AI researchers, and policymakers, the workshop will facilitate meaningful dialogue on designing AI systems that empower teachers and support students in an increasingly AI-driven educational landscape.

2 Workshop Themes and Topics

The workshop will center on two key themes, examining how AI can enhance teachers' roles in delivering personalized education while addressing the emotional and psychological dimensions of human-AI collaboration. These themes will shape discussions on the future of AI in education, focusing on developing systems that boost teaching effectiveness and foster empathy in learning environments.

2.1 Theme 1: AI-Powered Tools for Personalized Feedback and Enhanced Teaching

The first theme will examine how AI-driven educational tools can help teachers deliver more effective and personalized feedback across diverse subject areas, such as writing, discussion-based courses, and project-based learning. Tools like ITS and LLMs can automate routine tasks such as grading and providing feedback, freeing up teachers to concentrate on higher-level instructional goals, including fostering creativity, critical thinking, and deeper student engagement.

Key discussion points will include:

- AI systems for personalized learning, including Intelligent Tutoring Systems (ITS) and adaptive assessment platforms
- The use of large language models (LLMs) to provide real-time, tailored feedback in areas such as writing and debate

- Improving teacher effectiveness through AI-generated insights into student performance
- Challenges related to scalability, accuracy, and maintaining human oversight in AI-driven feedback systems

2.2 Theme 2: Building Empathy and Emotional Awareness in AI Systems

The second theme will explore how AI systems can be designed to go beyond mimicking human cognitive abilities to also understanding and responding to the emotional and psychological states of students. For AI to effectively augment teachers, it must recognize and respond to students' emotional cues, providing empathetic feedback that addresses their needs beyond academic achievement. Achieving this requires the integration of advanced emotion recognition and affective computing technologies into AI systems.

Key discussion points will include:

- Emotion recognition and sentiment analysis in AI-powered educational platforms
- Designing AI systems that exhibit empathy and human-like understanding
- Navigating ethical concerns related to AI's impact on students' emotional well-being
- Balancing AI's capacity to mimic human empathy while maintaining the integrity of the teacher-student relationship

3 Pre-Workshop Plans

3.1 Target Audience

This workshop is designed for a diverse audience of researchers, including:

- Educators and Teachers: Seeking to use AI tools to enhance personalized learning and boost student engagement.
- AI Researchers and Developers: Specializing in creating AI systems for education, with a focus on personalized feedback and emotional intelligence.
- Policymakers and Administrators: Responsible for decisions regarding AI adoption in education, with an emphasis on promoting equity and ethical standards.
- Education Technologists: Involved in integrating AI technologies within schools, universities, and other educational settings.

3.2 Recruitment

To reach a broad audience, we will promote the call for participation through the following channels:

- Academic mailing lists, including CHI-ANNOUNCEMENTS and AI in Education forums.
- Social media platforms (LinkedIn, Twitter) and the organizers' personal networks.
- Direct email invitations to researchers and educators within our network who are engaged in AI and education projects.

3.3 Paper Submission and Review Procedure

Participants are invited to submit position papers that address one or more of the workshop's key themes, with a focus on developing AI-empowered educational tools for personalized feedback or AI

systems that demonstrate human-like empathy in learning environments. Submissions should include the following:

- Length: Position papers should be no longer than 6 pages, excluding references, and follow the single-column format specified by the CHI Extended Abstract guidelines.
- Content: Papers should clearly identify the problem being addressed, describe the methodology or framework used, present key findings or insights, and explain how the work relates to the workshop's central themes.
- Motivation Statement: Authors are encouraged to include a brief statement outlining their interest in the workshop and how their research contributes to the themes of AI-augmented teaching or empathy-driven AI systems in education.
- Submission Portal: Papers should be submitted via the designated workshop email by the specified deadline.

All submitted papers will undergo a double-blind peer-review process to ensure quality and fairness. After an initial check by the workshop organizers for adherence to formatting and relevance, each paper will be reviewed by at least two members of the program committee. Reviews will evaluate the papers based on their relevance to the workshop themes, originality, research quality, potential impact, and contribution to meaningful discussion. Reviewers' feedback will be provided, and the selection process will aim for a diverse range of perspectives, balancing theoretical and practical contributions. Authors will be notified of acceptance and receive feedback for further refinement.

3.4 Workshop Format

The workshop will be structured to foster maximum interaction, collaboration, and engagement among participants. It will feature a blend of synchronous in-person activities and opportunities for asynchronous contributions, ensuring flexibility to accommodate the diverse needs of attendees.

3.4.1 Pre-Workshop Preparation. To encourage meaningful discussions, participants will be asked to upload a brief introduction video or presentation about their work and interests to the workshop website two weeks prior to the event. This will enable attendees to familiarize themselves with each other's research and identify shared interests in advance. Accepted position papers will also be posted on the workshop website for participants to review before the workshop.

3.4.2 Workshop Day Structure.

- In-Person Participation: The workshop will be a full-day event featuring presentations, group discussions, and collaborative activities, designed to foster deep engagement with the workshop themes and promote interaction among participants.
- Accessibility Features: To ensure inclusivity, all presentations and discussions will include captioning, and participants will be encouraged to inform organizers of any specific accommodation needs.

Key components of the workshop format include:

Table 1: Workshop Schedule

Time(JST)	Workshop Activity
09:00 am	Welcome
09:10 am	Keynote Presentation
09:50 am	Morning Tea
10:00 pm	Position Paper Presentations
12:00 pm	Lunch
01:30 pm	Group Activity
04:00 pm	Coffee Break
04:30 pm	Group Presentations and Discussion
05:30 pm	Workshop Wrap-up

- **Position Paper Presentations:** Participants will briefly present their accepted papers, followed by a Q&A session for deeper discussion and constructive feedback.
- **Group Activities:** Collaborative group exercises will focus on key themes, such as developing AI tools for personalized learning and integrating empathetic AI into educational settings.
- **Panel Discussions:** Expert-led panels on AI and education will provide opportunities to explore broader ethical, policy, and technical issues related to human-AI collaboration in classrooms.

The day's structure will balance theoretical discussions with practical explorations, encouraging participants to share insights and lay the groundwork for ongoing collaboration.

4 Workshop Structure

The workshop aims to promote rich interaction and collaboration, providing participants with opportunities to explore the themes of AI-augmented teaching and empathetic AI systems in education. The day will feature a balanced mix of formal presentations, hands-on group activities, and open discussions to ensure an engaging and well-rounded experience. The schedule is thoughtfully designed to facilitate deep exploration of the topics while generating actionable insights. Table 1 presents a preliminary schedule, which is subject to change.

5 Post-Workshop Plans

The main goal of this workshop is to promote meaningful collaboration among educators, researchers, and technologists in exploring AI's potential to augment teaching, enhance personalized learning, and develop emotionally intelligent AI systems. By the conclusion of the workshop, participants will have gained deeper insights into the current and emerging capabilities of AI in education and established connections with leading experts. These connections will pave the way for continued research, publications, and innovative projects in the field.

5.1 Academic Publications and Special Issue

A key outcome of the workshop will be the publication of a workshop proceedings document, which will include all accepted position papers and key insights from the group activities. To maximize

accessibility and impact, the proceedings will be published on a widely recognized open-access platform such as CEUR Workshop Proceedings.

To further promote scholarly dissemination, we plan to organize a special issue in a leading journal focused on AI in education, such as the International Journal of Artificial Intelligence in Education (IJAIED) or Computers & Education. Authors of exceptional papers will be invited to submit extended versions of their work for peer review, allowing them to refine their research based on the feedback received during the workshop. This special issue will feature cutting-edge advancements in AI-augmented teaching, personalized learning tools, and the integration of empathy in AI systems.

5.2 Knowledge Exchange and Continued Collaboration

In addition to academic publications, we aim to build a dynamic community of practice. A Slack group will be established to facilitate ongoing discussions, resource sharing, and collaborative opportunities among participants. This platform will enable researchers to exchange ideas, share project updates, and explore potential partnerships, fostering a global network of experts dedicated to advancing AI in education.

The workshop website will serve as a long-term resource hub, hosting the proceedings, video recordings of key presentations, presentation slides, and relevant publications. This repository will be accessible to all participants and the wider academic community, ensuring that the knowledge shared during the workshop continues to inspire future research and innovation.

5.3 Report and Future Initiatives

After the workshop, we will compile a comprehensive report summarizing the discussions, key insights, and outcomes. This report will be shared with participants and made publicly available on the workshop website, featuring practical recommendations for developing AI systems that enhance teachers' capabilities and promote emotionally intelligent learning environments.

We also plan to create a "Manifesto on AI-Augmented Teaching," which will outline key principles and challenges for the responsible integration of AI in education. This manifesto will serve as a guiding document for researchers, educators, and policymakers, advocating for AI systems that complement human teachers while upholding ethical standards.

Participation in this workshop will provide attendees with valuable insights into the latest AI applications in education, opportunities to contribute to impactful academic publications, and the chance to join a global network of innovators in this rapidly advancing field.

6 Call For Participation

AI has the potential to transform education by enhancing teachers' ability to deliver personalized instruction and create emotionally intelligent learning environments. Our workshop, "Augmented Educators and AI: Shaping the Future of Human-AI Collaboration in Learning," explores how AI can support teachers and how AI systems can be designed to demonstrate human-like empathy. We

invite educators, researchers, and technologists to join us in discussing the future of AI-augmented education.

The workshop will focus on two key themes: (1) developing AI tools that empower teachers to provide personalized feedback and tailored learning experiences, and (2) designing AI systems capable of understanding and responding to students' emotional and psychological needs, fostering human-centered interactions in educational settings. By addressing these themes, we will explore the opportunities and challenges of integrating AI into classrooms, considering ethical implications, technical advancements, and practical applications.

We welcome position papers that address one or both themes. Submissions may present original research, case studies, or theoretical perspectives on AI's role in enhancing education. Papers should outline research goals and problems addressed, limited to 6 pages (excluding references), following the CHI Extended Publication format.

Email submissions to kimhyejee923@seoultech.ac.kr by February 13, 2025, the Submission Deadline. For submission guidelines and details, visit our workshop website or contact the organizers (<https://sites.google.com/view/augmented-educators-and-ai/home>). All submissions will be reviewed by experts, with selection based on relevance, originality, and impact. Authors of accepted papers must attend and present their work. Participants must register for both the workshop and at least one day of the conference.

7 Organizers

The workshop is organized by a diverse team of international scholars, combining the expertise of established researchers. With backgrounds in HCI, AI in education, and personalized learning, they share a commitment to advancing AI's role in supporting educators and promoting human-centered education. Their collective interests include AI-driven tools for personalized feedback, empathetic AI systems, and the ethical integration of AI in education.

Hye Ji Kim is a visiting assistant professor in AI Convergence and Open Sharing Systems at Seoul National University of Science and Technology. Her research focuses on AI-driven educational technologies, particularly enhancing personalized learning and student engagement. As an interdisciplinary researcher, she explores Sensible Engagement with AI in Education, examining how AI can facilitate meaningful human-AI interaction in a mutually supportive framework. Her work emphasizes AI systems that complement, rather than replace, human educators, with a focus on addressing the ethical implications of AI integration in education.

Jongyoul Park is an associate professor at Seoul National University of Science and Technology. His research spans from real-time large-scale image understanding to AI-based human pose estimation and behavior recognition in surveillance systems. He has pioneered advancements in reinforcement learning, AI for visual big data analysis, and real-world applications like automated object detection and video interpretation. His work, recognized through numerous publications and patents, centers on leveraging AI to enhance precision in visual recognition systems, while addressing key challenges in AI integration and ethical considerations.

Hyeongbae Jeon is the Chief Research Officer at Tutoruslabs, where he focuses on developing personalized tutoring systems

powered by large language models. With a background in speech recognition and AI, he specializes in leveraging advanced AI technologies to create adaptive learning environments that cater to individual student needs. His work emphasizes the integration of language models to enhance the interactivity and effectiveness of AI-driven tutoring systems, aiming to revolutionize personalized education through innovative AI solutions.

Sidney Fels is a full professor in Electrical and Computer Engineering at the University of British Columbia, where he leads the Human Communication Technologies Lab. His research spans Human-Computer Interaction, virtual reality, biomechanical modeling, and machine learning, with a focus on enhancing human communication through advanced technology. He has made significant contributions to virtual reality, including the development of pioneering products like GesturePlus and CyberServer. With over 350 publications and multiple ongoing projects funded by prestigious grants, he collaborates across disciplines to push the boundaries of virtual reality in measuring cognitive abilities and transforming human interaction.

Samuel Dodson is an assistant professor in the Department of Information Science at the University at Buffalo, specializing in how scientists and engineers interact with information across various genres, such as regulatory documents and clinical reports. His interdisciplinary research, grounded in genre studies and activity theory, aims to illuminate document practices that shape scientific inquiry and technological problem-solving. With a background in Computer Science, Psychology, and Library Studies, Samuel is also dedicated to teaching courses on human-computer interaction, information visualization, and research methods, empowering students to navigate the complexities of STEM education.

Kyoungwon Seo is an assistant professor in the Department of Applied Artificial Intelligence at Seoul National University of Science and Technology. As an interdisciplinary researcher, he combines expertise in Human-Computer Interaction (HCI), Augmented/Virtual Reality, Cognitive Psychology, and Artificial Intelligence (AI). His work centers on creating human-centered AI systems that deepen user understanding and amplify their capabilities through data science, HCI techniques, and machine learning methods. He has collaborated with a range of industry and academic partners—such as Microsoft product designers, Hyundai Motor Company engineers, medical experts, and educators—to tackle complex challenges. Passionate about bridging technology and human needs, he aims to shape a future where AI empowers people in meaningful ways.

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